Dora Elalaoui-Pinedo

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My research interests center on surface processes across Mars, Earth, and Europa — including geomorphology, mineralogy, and the evolution of ice sheets and glaciers. I am interested in using imaging spectroscopy, remote sensing, field geology, and change detection techniques, along with data science and machine learning methods, to study active planetary surfaces and assess planetary habitability. I am also drawn to planetary science mission instrumentation.

EDUCATION

University of Arizona — Tucson, AZ
Department of Geosciences, Lunar & Planetary Laboratory, Department of Mathematics — Expected May 2027

- B.S. Planetary Geosciences, B.S. Data Science with an Applied Statistics Emphasis
- Minors in Astrobiology, Leadership Studies & Practice

Relevant Courses: Mineralogy, Petrology, Structural Geology, Mars, Intro to Statistical Computing
Cumulative GPA: 3.70

RESEARCH & WORK EXPERIENCE

University of Arizona, Undergraduate Student Researcher — Tucson, AZ
Aug. 2026–Present

- Produce digital terrain models of the South Polar Layered Deposits (SPLD) of Mars to catalog exposed strata, under Dr. Shane Byrne.
- Support the efforts to constrain the recent climate expressed in the geomorphic and geological features of the SPLD.

University of Arizona, HiRISE Student Research Technician — Tucson, AZ
May 2024–Present

- Create digital terrain models (DTMs) of the surface of Mars with stereophotogrammetry.
- Collaborate with researchers to explore geological processes on Mars using models.
- Teach and support attendees in a biannual planetary photogrammetry workshop.

SETI Institute, REU Intern — Mountain View, CA
Characterizing Morphological and Mineralogical Variability of Interdunes at Olympia Undae, Mars — Jun. 2026–Aug. 2026

- Analyzed morphological and mineralogical patterns in Martian interdunes using CRISM and HiRISE data, under Dr. Janice Bishop and Dr. Lori Fenton.
- Mapped interdune morphology using JMARS and collected mineral spectra with ENVI.
- Identified spectral evidence of iron sulfates, including potential rozenite, in interdunes.
- Investigated spectral evidence of evaporite minerals in interdunes to uncover aqueous processes shaping Mars's north polar climate history.

University of Arizona, NASA Space Grant Undergraduate Intern and Mentor — Tucson, AZ
Mapping Enigmatic Pits in the North Polar Layered Deposits (NPLD) of Mars — Aug. 2023–May 2025

- Investigated meter-scale pits using HiRISE imagery under Dr. Sarah Sutton.
- Analyzed more than 400 HiRISE images and located and mapped over 2,000 pits using QGIS.
- Created DTMs to conduct a temporal analysis of potential pit morphology changes.
- Identified evidence of active surface processes, potential faults, and ice sublimation or deposition in the NPLD to help uncover Mars's climate history.
- Mentored first-year Space Grant Interns through the research process and created networking experiences.

PUBLICATIONS

1. **Elalaoui-Pinedo, D.E.**, Sutton, S.S., et al. (in preparation). Enigmatic Pits in the North Polar Layered Deposits of Mars: Mapping and Seasonal Analysis.
2. Kelly, E., Andrews, S., [...] **Elalaoui-Pinedo, D.E.**, et al. (submitted). HCN and HNC 1–0 Emission in the Central Taurus Molecular Cloud. *Publications of the Astronomical Society of the Pacific.*

CONFERENCE PRESENTATIONS

(* = Oral)

1. ***Elalaoui-Pinedo, D.E.**, Bishop, J.L., Fenton, L.K., et al. (2026). Analyzing Patterns in Morphology and Mineralogy of Interdunes at Olympia Undae, Mars. *Geological Society of America Connects 2026*, Abstract #20070.

2. **Elalaoui-Pinedo, D.E.**, Sutton, S.S., Okubo, C., Byrne, S. (2025). Investigating Enigmatic Pits in the North Polar Layered Deposits of Mars. *Lunar and Planetary Science Conference LVI*, Abstract #2486.
3. ***Elalaoui-Pinedo, D.E.** & Sutton, S.S. (2025). Investigating Enigmatic Pits in the North Polar Layered Deposits of Mars. *Arizona NASA Space Grant Statewide Student Research Symposium XXXIV*.
4. ***Elalaoui-Pinedo, D.E.** & Sutton, S.S. (2024). Mapping Enigmatic Pits in the North Polar Layered Deposits of Mars. *Arizona NASA Space Grant Statewide Student Research Symposium XXXIII*.

TECHNICAL SKILLS & LANGUAGES

- Python, R, MATLAB, QGIS, SOCET SET, ENVI/IDL, JMARS, LaTeX, Microsoft Office (Word, Excel, PowerPoint, etc.), remote sensing, spectral analysis, image analysis, data analysis, scientific writing.
- Native in English and Spanish.

LEADERSHIP & TEACHING EXPERIENCE

Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS), Tucson, AZ
***President,* University of Arizona** Apr. 2024–Present

- Organize science education outreach at underserved Tucson public schools, leading 10 visits since 2024.
- Direct monthly community-building events and professional development programming.
- Guide chapter meetings and connect with other campus organizations to grow member engagement and participation.
- Lead and participate in environmental conservation volunteer efforts alongside local advocacy groups.

Housing and Residential Life Tucson, AZ
***Resident Assistant,* University of Arizona** Jul. 2023–May 2026

- Served as a mentor for ~30 residents, fostering community through regular one-on-one check-ins and dorm-wide events on topics such as academic success and diversity.
- Connected residents to campus resources to help them navigate challenges during their time as a student.
- Responded to and de-escalated resident conflicts, crises, and policy violations to ensure resident safety and wellbeing.

Blue Chip Leadership Tucson, AZ
***Peer Mentor,* University of Arizona** Aug. 2024–May 2025

- Facilitated a leadership development curriculum for a cohort of 20 first-year students, covering topics such as goal setting, leadership styles, and community engagement.
- Mentored first-year students through academic and personal challenges, connecting them to campus resources and helping them build self-confidence.
- Led weekly group sessions and community service projects to apply lessons on identifying systemic obstacles to opportunity and translating them into actionable, student-led initiatives.

Invited Guest Speaker Sep. 2026
***Topics in Planetary Science (PTYS 395B),* University of Arizona**

- Delivered final SETI Institute REU presentation, "Characterizing Morphological and Mineralogical Variability of Interdunes at Olympia Undae, Mars."
- Facilitated a discussion and Q&A session with students regarding the REU application process, timeline, and experience.

HONORS & AWARDS

Facilitated Graduate Admissions Program (F-GAP Scholar), Math Alliance at the American Mathematical Society 2026-2027
William Y. Vélez Scholarship 2026-2027
Hispanic Alumni Tuition Scholarship 2022-2026
Joel Valdez Scholarship 2022-2026

PROFESSIONAL ORGANIZATIONS & GROUPS

GeoLatinas 2026
Geological Society of America 2026
Epsilon Eta Pre-Professional Environmental Fraternity, Tau Chapter 2025
Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS) 2022